INCH POUND

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19 March 1973

PERFORMANCE SPECIFICATION

ANTENNA AS-1304( )/ARN-30

This specification is approved for use by the USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of Very High Frequency (VHF) Omnidirectional range balanced loop airborne antenna for the reception of Instrument Landing System (ILS), Localizer and Very High Frequency Omni-range (VOR) signals for use with Airborne Navigational equipments within the frequency range of 108 to 122 MHz and designated AS-1304( )/ARN-30 (see 6.1 and 6.3).

1.2 Appendices. The appendix listed below forms a part of this specification. Appendix - Temperature Altitude Test.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A: FSC 5826

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-E-5400	-	Electronic Equipment, Aerospace, General Specification for
MIL-M-13231	-	Marking of Electronic Items
MIL-F-14072	-	Finishes for Ground Signal Equipment

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810	-	Environmental Test Methods

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454	-	General guidelines for military equipment

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

COMMUNICATIONS-ELECTRONICS COMMAND

DL-SM-A-498900 - Antenna Assembly AS-1304( )/ARN-30

(Copies of these drawings are available from Commander, U.S. Army Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
AMERICAN SOCIETY for QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article samples. The contractor shall furnish three preproduction samples for approval (see 4.2 and 6.2).

3.2 Construction. Design and construction of the antenna shall be the contractor's responsibility. However, the interface and dimensional requirements of DL-SM-A-498900 shall be met.

3.3 Cleaning.

3.3.1 Parts. After fabrication, parts shall be cleaned. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.3.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.4 Finish, protective. The finish shall be suitable for the environment in which the antenna will be used. Drawing DL-SM-A-498900 and MIL-F-14072 may be used as guides. The use of Film B on type I surfaces for the final paint film is recommended.

3.5 Marking.

3.5.1 General. Mark directly or by label to include nomenclature, contract number, serial number, date, manufacturing code, part number, and national stock number. Use clear and legible lettering and a method of marking to withstand the operating environment. DL-SM-A-498900 and MIL-M-13231 may be used as guides.

3.5.2 Visibility. Whenever practicable, parts shall be to mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial Numbers. The unit(s) on this order shall have serial numbers.

3.6 Tropicalization of materials. Materials shall be treated as follows:

3.6.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.6.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.6.3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.7 Performance characteristics.

3.7.1 Output impedance (see 4.5.1). The unit(s) feed ports shall have a 50 ohm output impedance from 108 to 122 MHz.

3.7.2 Voltage standing wave ratio (VSWR) (see 4.5.2). The impedance of the antenna at the input connector of the cable assembly, shall be such that the VSWR referenced to a 50 ohm system does not exceed 5.0 to 1 throughout the frequency range of the antenna. The measurement shall be made with the antenna mounted on the ground plane of Figure 3.

3.7.3 Polarization (see 4.5.3). The antenna shall receive horizontally polarized signals. Reception of vertically polarized signals shall be at least 12 dB down from the level of

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horizontally polarized signals. The measurement shall be made with the antenna mounted on the ground plane of Figure 3.

3.7.4 Radiation pattern (see 4.5.3). The radiation field pattern shall be essentially omnidirectional in the azimuth plane and shall approximate a Cos θ curve in the vertical plane.

3.7.5 Gain (see 4.5.4). Antenna gain relative to an isotropic source shall not be less than -2 dB.

3.8 Operational inspection. Each unit(s) shall be capable of meeting the specified test after applicable processes inspection and test have been completed to determine that each unit(s) on order is operable prior to packaging (see 4.9).

3.9 Interchangeability (see 4.8). Like units, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Components of the antenna shall not be designed as matched units. Antenna half loops shall be completely interchangeable and shall perform as indicated in 3.8 and interface with all cable assemblies in DL-SM-A-498900.

3.10 Environmental conditions (see 4.6). The unit(s) shall meet the following environmental conditions.

3.10.1 Temperature-altitude (see 4.6.1). Antenna performance shall be as indicated for ambient temperatures from -54°C to +95°C at altitudes from sea level to 70,000 feet. The antenna shall withstand extended exposure to temperatures within the -62°C to +71°C range.

3.10.2 Humidity (see 4.6.2). The antenna shall withstand relative humidity conditions in excess of 95% and shall be capable of satisfactory performance, within 1/2 hour after subjection to the test.

3.10.3 Vibration (see 4.6.3). The antenna shall perform satisfactorily after subjection to vibration environments in accordance with the test of 4.6.3.

3.10.4 Shock (see 4.6.4). The antenna shall perform satisfactorily after subjection to a shock environment in accordance with the shock test of 4.6.4.

3.10.5 Acceleration (see 4.6.5). The unit(s) shall withstand 6g steady state acceleration for one minute.

3.10.6 Rain (see 4.6.6). The unit(s) shall meet the performance requirements of this document when subjected to rain as encountered in service, with the base down, in accordance with the rain test (see 4.6.6).

3.10.7 Salt fog (see 4.6.7). The unit(s) shall be exposed, in normal operation configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by

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any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.10.8 Sand and dust (see 4.6.8). The unit(s) shall withstand exposure to sand and dust particles as encountered in desert areas. It shall show no evidence of corrosion as evidenced by any visible degradation of material and finish.

4. VERIFICATION

4.1 Classification of inspection. Inspection shall be classified as follows.

a. First Article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article inspection shall be performed by the contractor. It shall consist of the inspection specified for Group A, Group B, and Group C (see Tables I, II and III respectively).

4.3 Conformance inspection. The contractor shall perform the inspection specified in 4.3.1 through 4.3.4. This does not relieve the contractor of his responsibility for performance of any additional inspection which is necessary to control the conformance of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.7).

4.3.1 Group A Inspection. Each unit shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance

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with ANSI/ASQCZ1.4. Each lot shall be subject to sampling inspection, utilizing the procedures of ANSI/ASQCZ1.4.

TABLE I. Group A inspection

Inspection	Requirement paragraph	Inspection paragraph
Visual and mechanical AS-1304( )/ARN-30: Base assemblies (less top) Base assemblies (assembled) AS-1804( )/ARN-30: (assembled)	3.1 - 3.5	4.7
Electrical:		
Voltage standing wave ratio	3.7.2	4.5.2
Polarization	3.7.3	4.5.3
Operational inspection	3.8	4.9

4.3.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met-the Group A inspection. This inspection shall conform to Table II.

TABLE II. Group B inspection

Inspection	Requirement paragraph	Inspection paragraph
Output impedance	3.7.1	4.5.1
Interchangeability	3.9	4.8

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4.3.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table III. Samples shall be selected in accordance with 4.3.3.1.

TABLE III. Group C inspection

Inspection	Requirement paragraph	Inspection paragraph
Subgroup I:		
Temperature/altitude	3.10.1	4.6.1
Humidity	3.10.2	4.6.2
Vibration	3.10.3	4.6.3
Radiation pattern	3.7.4	4.5.3
Gain, antenna	3.7.5	4.5.4
Subgroup II		
Shock	3.10.4	4.6.4
Acceleration	3.10.5	4.6.5
Rain	3.10.6	4.6.6
Salt fog	3.10.7	4.6.7
Sand and dust	3.10.8	4.6.8

4.3.3.1 Sampling for group C inspection. Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.3.3.1.1 Subgroup I. For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units each month shall be selected.

4.3.3.1.2 Subgroup II. For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units every two months shall be selected.

4.3.3.2 Order of inspection within group C. Group C testing shall be performed in accordance with Table IV in the sequence shown. The items shall be cleaned, dried and refurbished if required before proceeding to the next test.

TABLE IV. Group C inspection

SUBGROUP I	SUBGROUP II
Temperature/altitude	Rain
Radiation pattern	Sand and dust
Gain, antenna	Salt fog
Humidity	Acceleration
Vibration	Shock

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4.3.4 Reinspection of conforming group C sample units. Unless otherwise specified, simple units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.4 Standard test conditions. Unless otherwise specified, tests shall be performed under prevailing ambient conditions of temperature, pressure and relative humidity. The following test conditions shall apply when specified herein:

a. Cylindrical ground plane (see Figure 3).

(1) Material: Aluminum

(2) Thickness: 1/8 inch.

(3) Width: Approximately 21 inches

(4) Length: Approximately 132 inches

b. Load: 50 ohms

4.5 Performance inspection.

4.5.1 Output impedance. Set the Hewlett-Packard (H.P.) 803AVHF bridge or equal test meter to the required frequency and make required adjustments. Connect the test meter to the feed port. Measurements shall be made to determine compliance with 3.7.1.

4.5.2 Voltage standing wave ratio (VSWR) test. VSWR measurements shall be made under standard conditions. The VSWR shall be measured with the antenna installed on the ground plane of Figure 3. Measurements shall be made in increments of 2 MHz throughout the frequency range. After setting the meter to the required frequency, calculate the VSWR on a "Z-θ" chart for compliance with 3.7.2.

4.5.3 Pattern measurements. Radiation pattern measurements shall be made for both horizontal and vertical polarization of the source antenna. Measurements shall be made with the antenna mounted on the ground plane of Figure 3 at increments of 2 MHz throughout the frequency range (see 3.7.4).

4.5.4 Gain, antenna. The gain of the antenna shall be measured relative to an isotropic source. The antenna shall be tested to determine compliance with 3.7.5.

4.6 Environmental. The unit(s) shall be subjected to and meet the service conditions as Class 2 equipment, except as defined herein. The performance inspection of 4.5.2 shall be conducted prior to and subsequent to each service condition that requires a test to be conducted.

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The measurement in this test shall not change more than five (5%) percent from the value measured prior to subjection to the test condition; in no event shall the measurement exceed the specified limit. VSWR shall be measured during the temperature and humidity tests while the antenna is in chamber, in addition to tests prior and subsequent to those environmental tests. Any type of chamber or adequate size to hold the specified ground plane will suffice for these tests since only a possible change in the VSWR is looked for. A pre-test VSWR is to be made to obtain a chamber reference VSWR.

4.6.1 Temperature-altitude test. The temperature altitude test of the Appendix shall be performed. The antenna shall be non-operating during application of environmental conditions other than normal room ambient.

4.6.2 Humidity test. The following humidity test shall be performed. Subject a sample to natural cyclic high humidity conditions under randomly varied temperatures between 79°F and 95°C with relative humidity randomly varied between 79 and 100% over a 24 hour period. Stabilize the test item at ambient conditions for another 24 hours. A VSWR test in accordance with 4.5.2 shall be performed as soon as practicable after the completion of the test, but not later than 1/2 hour after completion (see 3.10.2).

4.6.3 Vibration test (see 3.11.3). The following vibration test shall be performed. MIL-STD-810, Method 514.1 may be used as a guide, but its use is not mandatory.

4.6.3.1 Procedure. Vibration amplitudes and frequencies shall be measured by techniques that will not significantly affect test item control or response. The input control sending device(s) shall be rigidly attached to the vibration table at, or as near as possible to the attachment point(s) of the test item.

a. Measure the VSWR in accordance with 4.5.2. Record the data.

b. The vibration shall be applied along each of the three mutually perpendicular axes of the test item. The vibratory acceleration levels or double amplitudes of the test curve in figure 4 shall be maintained at the test item mounting points.

c. Resonant frequencies of the test item shall be determined by varying the frequency of applied vibration slowly through the specified range at reduced test levels but with sufficient amplitude to excite the item. Sinusoidal resonance search may be performed using the test levels of figure 4. If the test levels of figure 4 are used, the time spent for this test can be counted as part of the total required test time.

d. The test item shall be vibrated along each axis at the most severe resonant frequencies determined in step c. Test levels and frequency ranges shall be in accordance with figure 4. If more than four significant resonant frequencies are found for any one dwell time at each resonance shall be 30 minutes. If a change in the resonant frequency occurs during the test, its time of occurrence shall be recorded, and immediately the frequency shall be adjusted to maintain the peak resonance condition. The final resonant frequency shall be recorded.

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e. The test item shall be vibrated along each axis in accordance with the test levels and frequencies of figure 4. The test time shall be three hours (less dwell time) along each of the three mutually perpendicular axes. The frequency of applied vibration shall be swept logarathmically over the range shown on figure 4. The sweep time from the lowest frequency on figure 4 to the highest frequency on figure 4 and then back to the lowest frequency on figure 4 shall be 36 minutes.

f. Remove the antenna from the vibration machine and install it on the ground plane of figure 3.

g. Perform the VSWR test in accordance with 4.5.2, and record the results. Compare the results with those obtained in step a. Any significant change in VSWR shall be considered to be a failure of the vibration test.

4.6.4 Shock test. The following equipment shock test MIL-STD-810, Method 516.1 may be used as a guide. At the completion of the test, the VSWR performance test of 4.5.2 shall be performed (see 3.10.4).

4.6.5 Acceleration test. The unit shall be tested in accordance with the following procedure. MIL-STD-810, Method 513.1 may be used as a guide. The contractor may, at his option, propose an alternate procedure to accomplish the same purpose.

a. Install the antenna on the ground plane described in figure 3.

b. Perform the VSWR test in accordance with 4.5.2, and record the data.

c. Remove the antenna from the ground plane.

d. Install the antenna on an appropriate centrifuge by its normal mounting means, with the long dimension along a radius of the centrifuge. The centrifuge rotational speed and the distance of the center of the antenna from the rotational center of the centrifuge shall be adjusted to provide an acceleration of 6G. If the end of the antenna nearest to the rotational center of the centrifuge has an acceleration of less than 90 percent of this G level, move the antenna farther out on the radius, and compensate by reducing the speed of rotation. If the end of the antenna farthest way from the rotational center of the centrifuge has an acceleration of more than 110 percent of this G level, move the antenna farther out on the radius, and compensate by reducing the speed of rotation.

e. Operate the centrifuge for one minute to provide an acceleration of 6G at the center of the antenna.

f. Repeat the test using the same acceleration level, along each of three mutually perpendicular axes in two opposite directions for a period of one minute in each direction for a total of 6 tests.

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g. Reinstall the test item on the ground plane.

h. Perform the VSWR test on the antenna, using 4.5.2, and record the data. Compare the data with the data taken in step 2. Any significant change in the VSWR shall constitute failure of the acceleration test.

4.6.6 Rain test. The antenna shall be mounted with base down for demonstrating compliance with 3.10.6. The equipment shall be subjected to the following test:

a. Place the antenna in the chamber and orient in such a manner as to allow impingement of the rain on the item normal (90°) to its horizontal plane.

b. Produce a simulated falling rain in the test chamber stabilized at a constant rate of 1 ± 0.5 inches per hour.

c. Reorient the test item to an angle of 45° from the horizontal plane of step 1 and index the item around its center axis periodically to allow exposure to rainfall of all the significant surfaces of the test item.

d. Expose the test item to the simulated falling rain for a total time of not less than 30 minutes, which shall represent the sum of equal exposure periods in the orientation modes of a and c.

e. At the conclusion of rain exposure, remove the test item from the chamber, sponge or lightly wipe accumulated moisture from its exterior surfaces, open its case, if possible, and visually inspect the test item.

f. At the completion of the test, the VSWR performance test of 4.5.2 shall be performed.

4.6.7 Salt fog test. The following salt fog test shall be performed. Salt Fog Test of MIL-STD-810, Method 509 may be used as a guide. Using spray apparatus, piping, and collector reservoir materials non-reactive to the salt spray, subject an antenna to a continuously atomized, finely divided, wet, dense salt spray mixture of 5% sodium chloride and 95% water. Uniformly distribute the salt fog over the test sample at a fallout rate between 0.3 and 3ml/80cm^2/hr for 48 hours at a constant 95°F with minimal air circulation. The antenna connector may be plugged to simulate installed condition. The unit shall not be required to operate electrically. The units shall meet the requirement of 3.10.7.

4.6.8 Sand and dust test. The unit(s) shall be subjected to the following sand and dust test. The Method 510 sand and dust test of MIL-STD-810 may be used as a guide. With the temperature at least 73°F with a relative humidity (maintained throughout the test) of no more than 30%, subject the antenna to a blowing sand and dust concentration (97 to 99% by weight) of fine sand and dust particles ranging from 100 to 325 microns at air velocities ranging form 300 to 1750 ft/min for at least 6 hours. Stop the test, adjust the temperature to 167°F, and continue test

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article exposure for another 6 hours. Stabilize at natural ambient conditions. At the completion of the test, the VSWR performance test of 4.5.2 shall be performed.

4.7 Visual and mechanical. The assemblies, prior to assembly, and the complete assembled unit(s) shall be examined for defects (see 6.2.f).

4.8 Interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of moldings, mounting bases, and insertable parts, when such dimensions affect mating.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Locations of holes on separable parts which must mate such as mounting assemblies.

d. Location of connectors, mounting base holes and elements.

e. Size and form of threads.

4.9 Operational inspection. Each unit shall be inspected for assurance that, when fed a modulated RF signal between 108 and 122 MHz and the demodulated signal is displayed on an oscilloscope, there shall be no interruptions or noise in the displayed signal as the unit(s) is vibrated at 5g's between 25 and 55 Hz (see 3.8).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended Use. The antenna is intended to be used with Airborne Radio Sets to provide a capability for both fixed and rotary wing aircraft, in conjunction with ILS Localizer and

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VOR Navigational radio equipment. The nomenclature AS-1304( )/ARN-30 has been retained because it is the nomenclature for AN/ARN-30( ) Radio Set Antennas presently in use.

6.2 Acquisition requirement. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level of preservations, packaging and level of packing required for shipment (see 5.1).

c. First article inspection. Three sample units of each item cited in Section are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on the third sample unit (see 3.1).

d. Marking and shipping of samples.

e. Place of final inspection.

f. Visual and mechanical defects.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AS-1304W/ARC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending results of the corrective action specified in the contract or purchase order.

6.5 Location of operational inspection. It is desirable that the operational inspection in 4.9 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive conformance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's conformance system and the history of the product (see 4.3).

6.8 Subject term (keyword) listing.

Gain
Output impedance
Polarization
Radiation pattern

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Voltage standng wave ratio (VSWR)

6.9 Changes from previous issue. Marginal notation are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian
Army - CR

Preparing Activity:
Army - CR

(Project 5826-A239)

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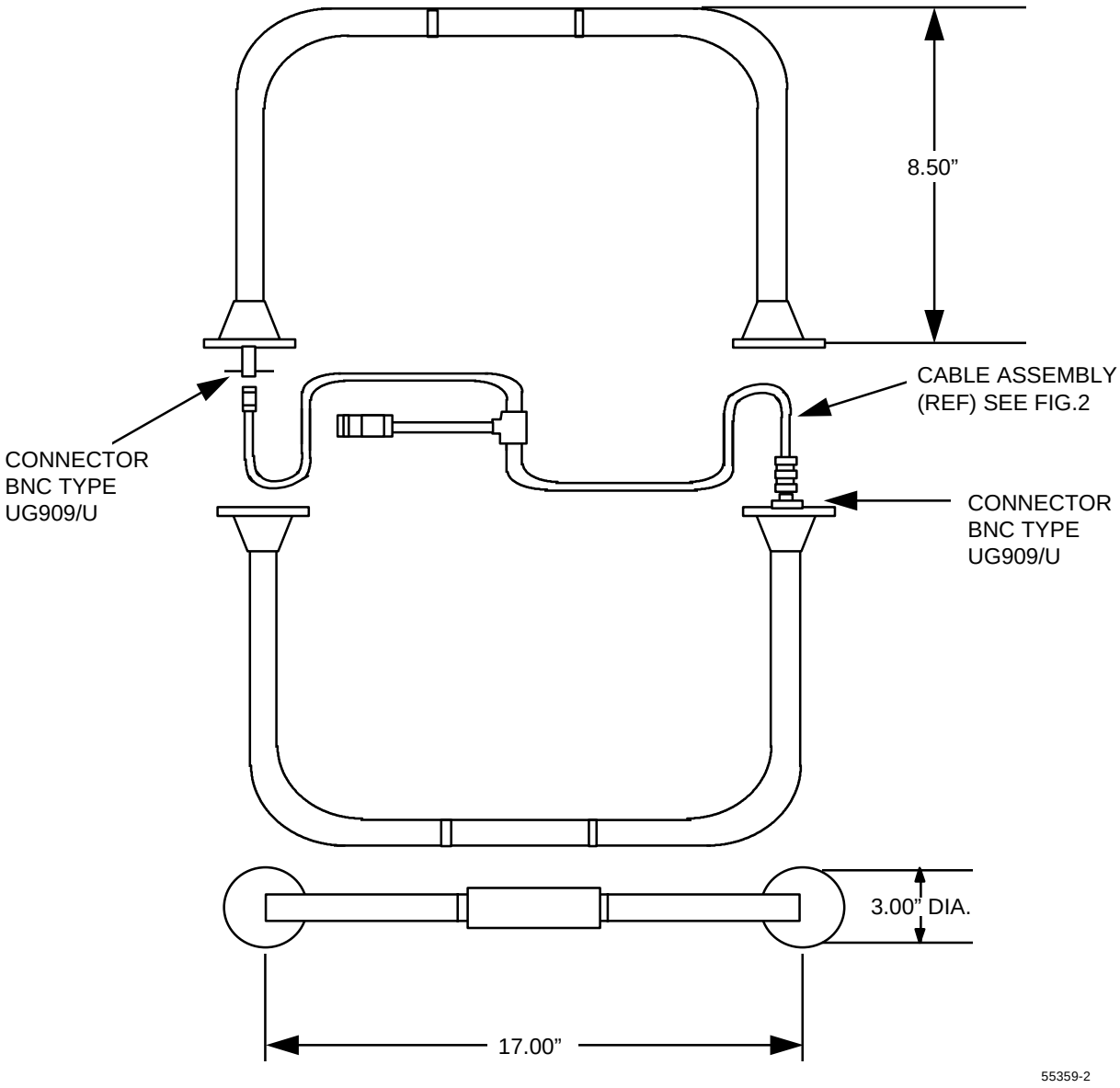
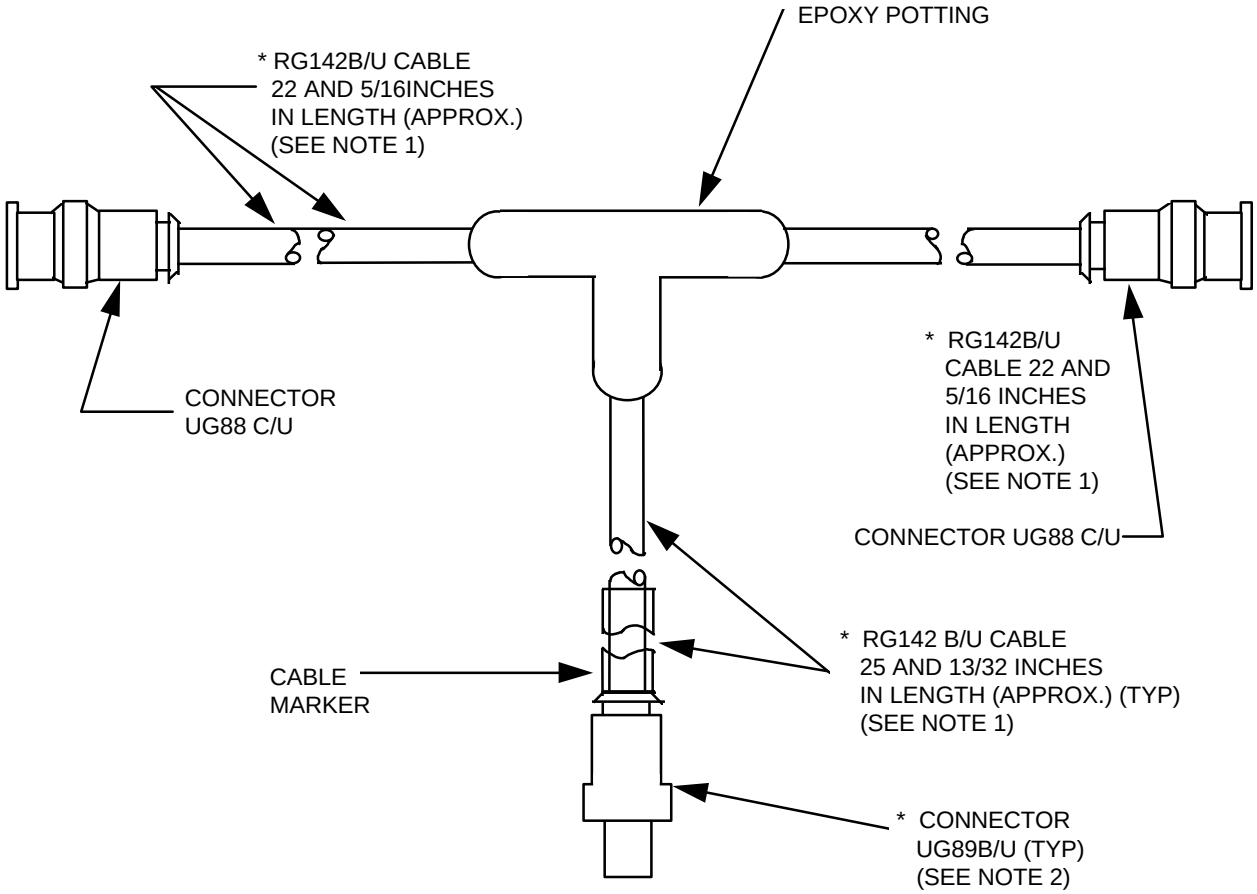


FIGURE 1. Antenna outline.

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* NOTES:

1. FOR EXACT DIMENSION REQUIRED SEE DRAWINGS.
2. FOR TYPE OF CONNECTOR REQUIRED SEE DRAWINGS.

FIGURE 2. Antenna cable assembly outline.

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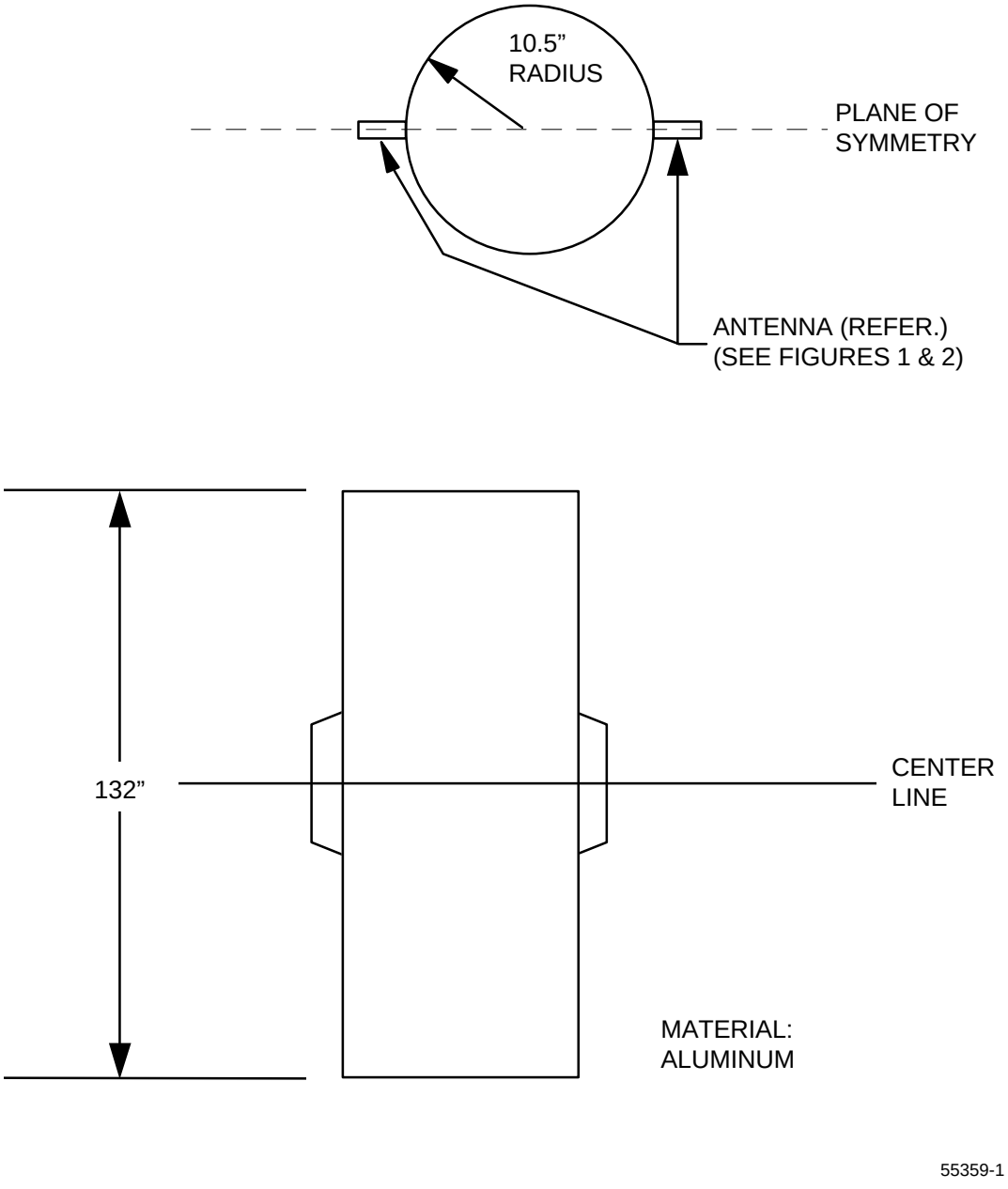


FIGURE 3. Ground plane.

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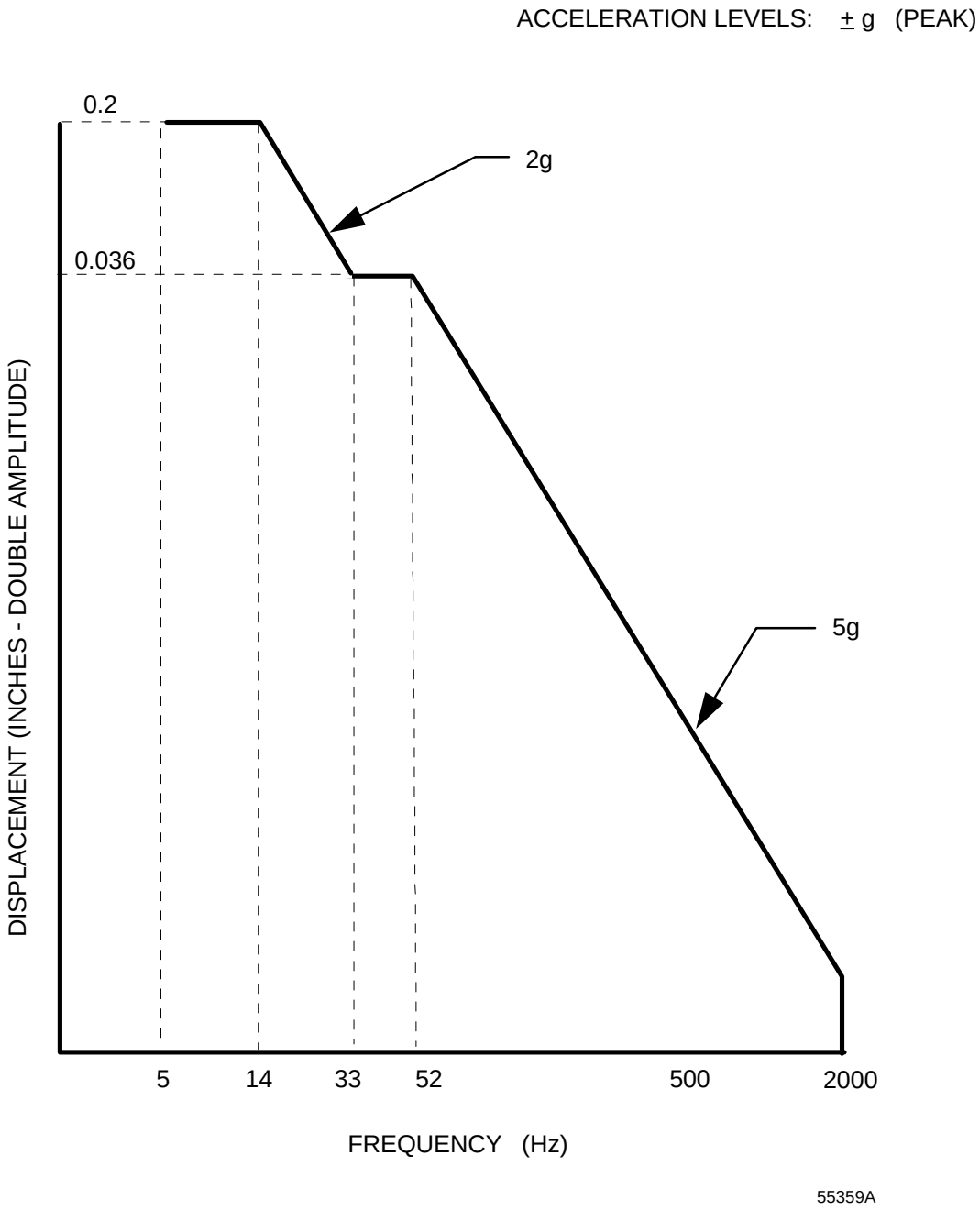


FIGURE 4. Vibration test profile.

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APPENDIX A

TEMPERATURE-ALTITUDE TEST

A.1 SCOPE

A.1.1 Scope. This appendix details the procedure for conducting a temperature-altitude verification. This appendix is a mandatory part of this specification.

A.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

A.3 PURPOSE. The temperature-altitude test is conducted to determine the ability of equipment to operate satisfactorily under simultaneously applied varying conditions of low pressure and high/low temperature.

A.4 PROCEDURE

A.4.1 General. In general, the testing schedule outlined in table A-II should be followed. When changing chamber conditions from those required for one step to those required for any other step, the rates of temperature and pressure changes may be the maximum attainable by the chamber, but these rates shall not exceed 1°C (1.8°F) per second and 0.5 inch of mercury per second for the AS-1304/ARN-30.

a. Install the test item on the ground plane described in figure 3.

b. Install the test item and the ground plane in the chamber.

c. Adjust the chamber for normal ambient temperature and pressure.

d. Perform the VSWR test of 4.5.2, and record the data. This will serve as the reference reading.

e. With no signal applied to the test item, adjust the chamber test conditions to those specified in step e of table II. After the test item temperature has stabilized, maintain these test conditions for at least the length of time specified in step c of table II. Where it is possible, without changing the test conditions, the test item shall be visually inspected for the presence of deterioration which would impair future operation.

f. With no signal applied to the test item, adjust the chamber test conditions to those specified in step f of table II. These chamber test conditions shall be maintained throughout this step. The above operational sequence shall be repeated two additional times.

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g. With no signal applied to the test item, adjust the chamber temperature to that specified in step g of table II. The test item temperature shall be stabilized and the chamber pressure adjusted to simulate the altitude specified in step g of table II.

h. With no signal applied to the test item, adjust the chamber test conditions to those specified in step h of table II. After the test item temperature has stabilized, the chamber door shall be opened, and frost permitted to form on the test item. The door shall remain open long enough for the frost to melt, but not long enough for the moisture to evaporate.

i. The chamber door shall be closed and the chamber temperature and pressure returned to normal room ambient. After the test item temperature has stabilized, an operational and performance check of the test item shall be made, utilizing the VSWR test of 4.5.2, and the results compared with the data obtained from the operational and performance tests performed in step d. The data shall be evaluated to determine whether any degradation had occurred.

j. With no signal applied to the test item, adjust the chamber test conditions to those specified in step j of table II. The test item temperature shall be stabilized and then maintained for at least the length of time specified in step j of table II. At the conclusion of this time period, where practicable, the test item shall be visually inspected for any deterioration.

k. With no signal applied to the test item, adjust the chamber test conditions to those specified in step k of table II and maintain until completion of the fourth operating time period. After the test item temperature has stabilized, the test item shall remain at this temperature for four time periods, each of the duration specified in step j of table II. Thermal sensor readings of the test item temperature shall be recorded every 10 minutes.

l. With no signal applied to the test item, adjust the chamber test conditions to those specified in step l of table II and maintain until completion of the fourth operating time period. After the test item temperature has stabilized, the test item shall remain under these conditions for four time periods each of the duration specified in step l of table II. Thermal sensor readings of the test item temperature shall be recorded at the beginning and end of each operating time period.

m. With no signal applied to the test item, adjust the chamber temperature to that specified in step m of table II. The test item temperature shall be stabilized. The chamber pressure shall be adjusted to simulate the altitude specified in step m of table II. Maintain these test conditions for the length of time specified in step m of table II. Thermal sensor readings of the test item temperature shall be recorded every 30 minutes.

n. With no signal applied to the test item, adjust the chamber temperature to that specified in step n of table II, and maintain until completion of the fourth operating time period. The test item temperature shall be stabilized. The chamber pressure shall be adjusted to simulate the altitude specified in step n of table II. The test item shall remain under these conditions for four time periods, each of the duration specified in step n of table II. Thermal sensor readings of the test item temperature shall be recorded every 10 minutes of test item operation.

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o. With no signal applied to the test item, adjust the chamber temperature to that specified in step o of table II. The test item temperature shall be stabilized. The chamber pressure shall be adjusted to simulate the altitude specified in step o of table II. Maintain these test conditions for the length of time specified in step o of table II. Thermal sensor readings of the test item temperature shall be recorded every 30 minutes.

p. Adjust the chamber test conditions to standard ambient conditions. After the test item temperature has stabilized, an operational and performance check of the test item, consisting of the VSWR test of 4.5.2, shall be made and the results compared with the data obtained in step d. Any significant change of VSWR shall be considered to be a failure of the temperature-altitude test.

TABLE I. Temperature-altitude test requirements.

Altitude range (ft)	Temperature (°C)	Equipment mode		
		operating		non-operating
		continuous	intermittent	
sea level to 70,000	-54 to 55	x	-	-
	55 to 71	-	x	-
	-54 to 95	-	-	x

TABLE II. Test conditions for temperature-altitude tests.

Step	c	d	e	f	g	h	i	j	k	l	m	n	o	p
Temp (°C)	*s	*s	-62	-54	-54	-10	*s	71	95	48	64	40	57	*s
Alt (ft)	*s	*s	site	site	30k	site	*s	site	site	50k	50k	70k	70k	*s
Time	*s	*s	2 hr	---	---	---	*s	20 hr	30 min	4 hr	30 min	4 hr	30 min	*s

*s standard conditions

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